

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

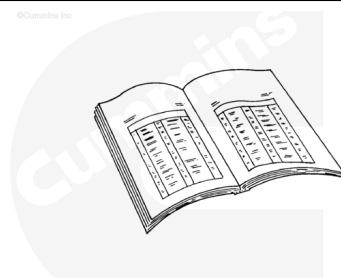
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Use caution when draining oil or replacing filters that oil is not spilled or drained into the bilge area. The oil and oil filters must be discarded in accordance with local environmental regulations.

- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-013.html)
- Drain the lubricating oil from engine. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/89/89-007-037.html)
- Remove the wiring harness coming from the engine control module (ECM) that may interfere with oil filter head removal. Use the following procedure in the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113. Refer to Procedure 019-043 in Section 19.
(/qs3/pubsys2/xml/en/procedures/19/19-019-043.html)

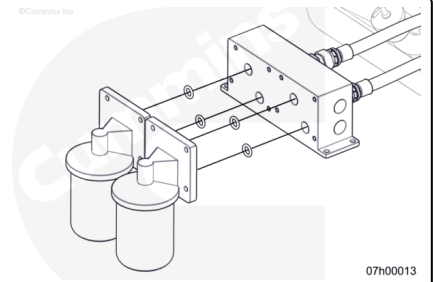


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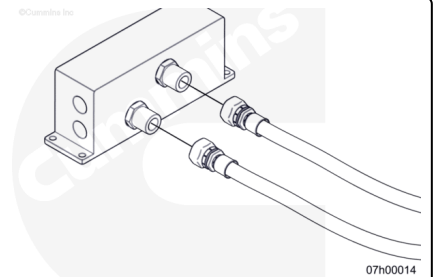
Remove

Remove the capscrews securing the filter heads to the remote mount adapter. Remove filter heads and discard o-rings. Refer to Procedure 007-015 in Section 7.

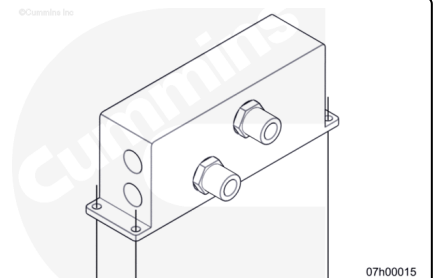
(/qs3/pubsys2/xml/en/procedures/89/89-007-015.html)



Disconnect the hoses from the back of the remote mount adapter.



Remove the capscrews and the remote mount adapter from the attachment base.



Remove the air compressor from the engine block and compressor brace.

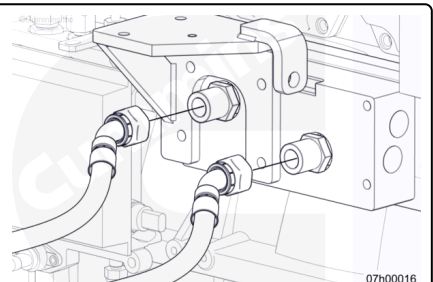
Refer to Procedure 012-014 in Section 12.

(/qs3/pubsys2/xml/en/procedures/89/89-012-014.html)

Note : There is no air compressor installed on power generation applications.

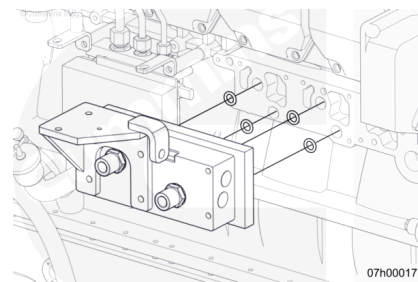


Disconnect the two 90 degree elbow hoses from the oil transfer block.



Remove the six capscrews, compressor brace, oil transfer block, and the o-rings from the engine clock. Discard the o-rings.

Note : On power generation applications there will be no compressor brace.

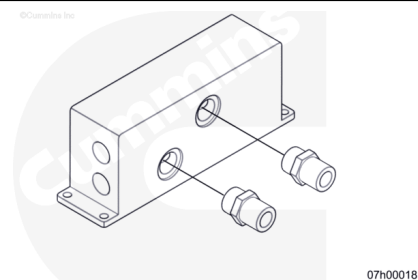


Disassemble

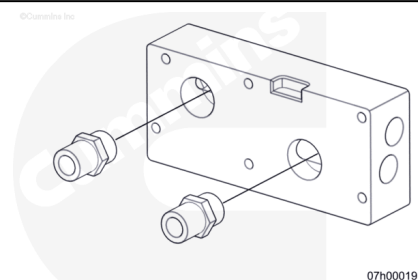
Note : Throughout this procedure it refers the user to place the components in a vise. The illustrations are without a vise for clarity.

Remove the two straight connectors from the back of the remote mount adapter while it is secured in a vise.

Note : If needed for cleaning, remove the two plugs from the remote mount adapter.



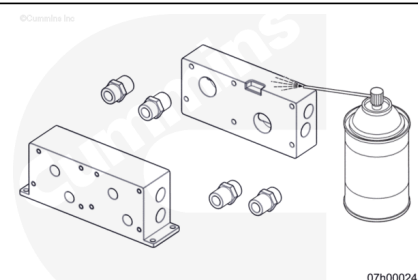
Remove the two straight connectors from the front of the oil transfer block while it is secured in a vise



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



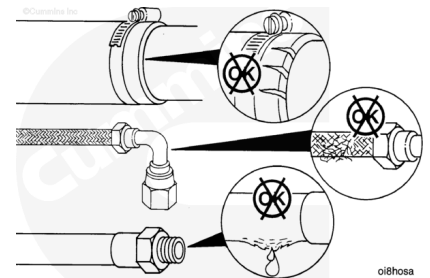
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Clean all parts with contact cleaner, Part Number 3824510, or equivalent.

Check lines and hoses for cracks.

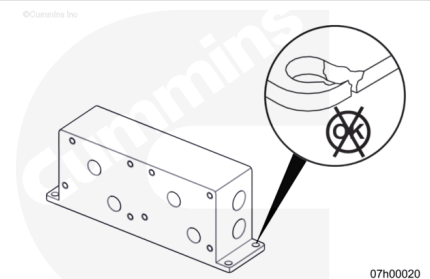
If the lines or hoses are cracked, they **must** be replaced.



Note : Remote mount lubricating oil filter head brackets may look different than in the illustration shown. The brackets are original equipment manufacturer (OEM) dependent.

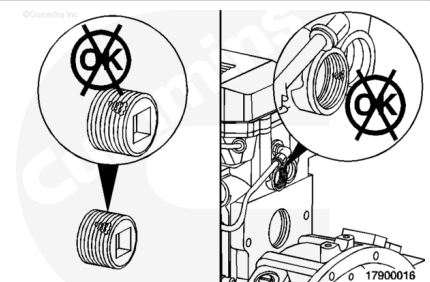
Inspect the lubricating oil filter head bracket for cracks and other damage.

If the lubricating oil filter head bracket is cracked or damaged, it **must** be replaced.



Inspect the threads of the pipe plugs and the fittings for damage. Replace if damaged.

Inspect the threaded bores for damage. Replace if damaged

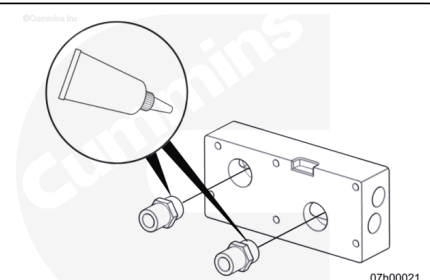


Assemble

Secure the oil transfer block in a vise.

Apply thread sealant, Cummins® Part Number 3375066, or equivalent, to the threads of the straight connectors to the front and rear of the oil transfer block.

Install the plugs into the oil transfer block, if removed.



Torque Value: 60 n•m [45 ft-lb]

Install the two straight connectors into the back of the oil transfer block.

Torque Value: 108 n•m [80 ft-lb]

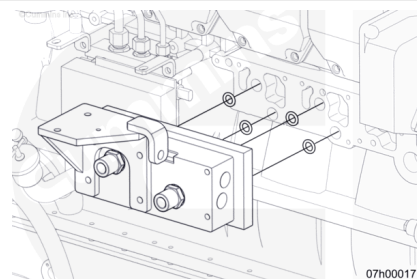
Install

Install the oil transfer block and four new o-rings to the engine block.

Install the compressor brace and six capscrews onto the oil transfer block. The four longer bolts are used for the air compressor brace. Tighten the capscrews.

Torque Value: 30 n•m [22 ft-lb]

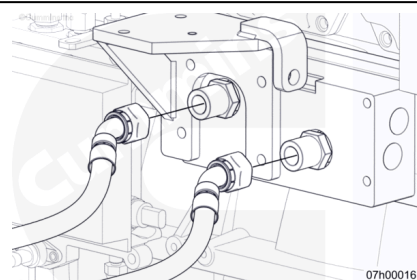
Note : For power generation applications there is no compressor brace. All capscrews used are the same length.



Connect the two 90 degree elbow hoses to the oil transfer block.

Torque Value: 176 n•m [130 ft-lb]

Make sure the oil lines do **not** interfere or make contact with other components on or off the engine.



Install the air compressor to the engine block and compressor brace.

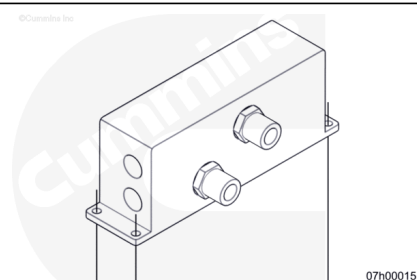
Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/89/89-012-014.html)

Note : There is no air compressor installed on power generation applications.



Install the remote mount adapter and capscrews to a mounting surface. Tighten the capscrews.

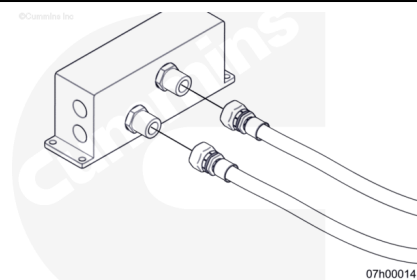
Torque Value: 55 n•m [41 ft-lb]



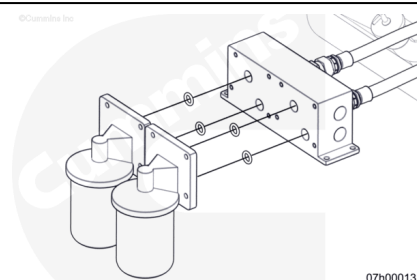
Install the hoses into the back of the remote mount adapter. The straight connector **must** be held in place with a second wrench while tightening the hoses.

Torque Value: 176 n•m [130 ft-lb]

Make sure the hose blocks correlate to the markings A-A and B-B from the oil transfer block to the remote mount adapter.



Install the oil filter mounts to the remote mount adapter using six new o-rings. Refer to Procedure 007-015 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-015.html>)



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the wiring harness coming from the ECM if it interfered with oil filter head removal. Use the following procedure in the QSK19, QSK23, QSK45, QSK60, and QSK78 Electronic Control System Troubleshooting and Repair Manual, Bulletin 3666113. Refer to Procedure 019-043 in Section 19. (</qs3/pubsys2/xml/en/procedures/19/19-019-043.html>)
- Install a new lubricating oil filter. Refer to Procedure 007-013 in Section 7 (</qs3/pubsys2/xml/en/procedures/89/89-007-013.html>).
- Fill the engine with clean 15W-40 lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/89/89-007-037.html>)

- Connect the battery. See equipment manufacturer service information.
- Start and operate the engine to check for leaks and proper oil pressure.
- Shut the engine off and allow 5 to 10 minutes for the oil to drain back to the oil pan. Check the lubricating oil level. Fill the sump to the high mark, if necessary.

Last Modified: 05-May-2015
